INCH-POUND

MIL-PRF-49471/8A(CR)
30 November 2000
Superseding
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PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X600/U

This specification is approved for use within Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-49471(CR).

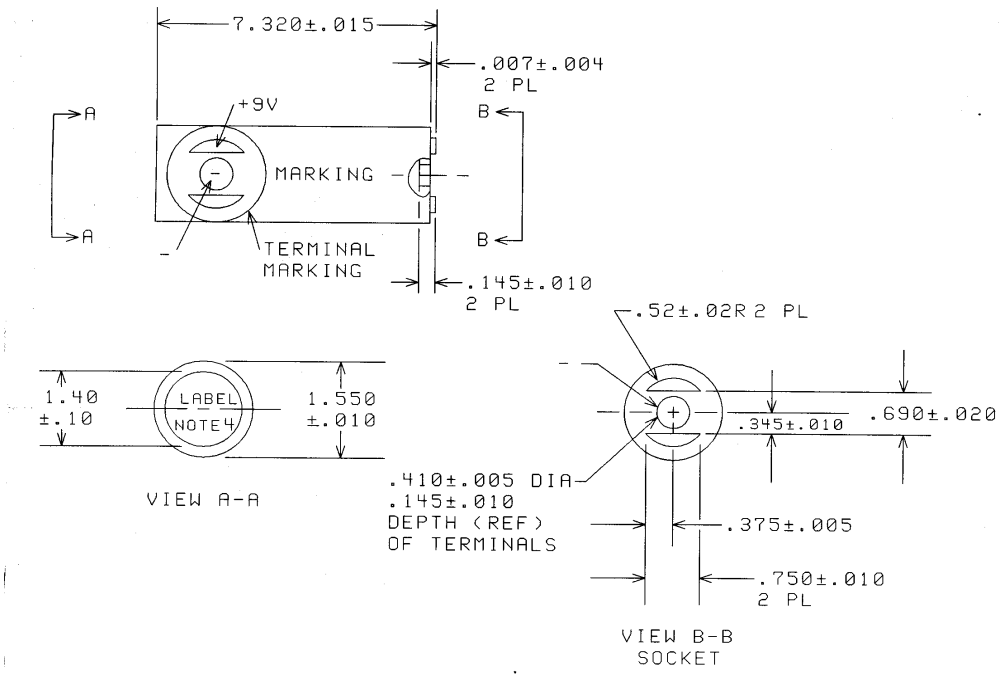

Figure 1. Battery dimensions and terminal markings.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:

1. All dimensions are in inches.

2. The battery terminals shall consist of two raised crescent-shaped (positive) terminals surrounding one central, recessed disc (negative) terminal. All terminals are integral with one end of battery (see Figure 1).

3. Adhesive decal shall be red/orange: Decal markings shall be in black letters (3/32 inch high) "This End Out", "Date Installed". This label need not be applied to First Article test samples.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 9.90 volts.

Cut-off voltage for capacity: 6.00 volts.

Terminals: Two (2) positive and one (1) negative terminals, all at one end of battery (see note 2).

Weight (maximum): 1.2 pounds (544 grams).

Battery short circuit test: Applicable.

Charge protection: Applicable.

High temperature protection: Applicable.

Complete discharge device: Applicable.

State of Charge Indicator: Not applicable.

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Minimum Capacity Requirement in Hours to 6.00 Volts
I	3.2
L	1.8
H	5.6
HT	5.4
LT	1.6
IT	2.7
LR1	47.0

Initial voltage delay: During the I, H, HT, and IT discharge tests, initial voltages below 5.70 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests initial voltages below 5.70 volts cannot exceed a 60.0 second duration.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all capacity tests are specified in basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 2.0 amperes at 70±5°F to 2.0 volts. Each discharged cell shall then be connected in series to two fresh cells. Each string shall be discharged at 2.0 amperes at 70±5°F for 3.2 hours.

Batteries: For all capacity tests, except the LR1, H and HT, the batteries shall be discharged at 2.0 amperes constant current to cut-off volts (zero volts for the I test). For the H and HT capacity tests the battery shall be discharged through 2.0 amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until closed-circuit voltage reaches cut-off volts. For the LR1 test, batteries shall be discharged at 135 milliamperes to 6.5 volts, followed by 1.7 watts to 1 volt.

Forced Discharge: After discharge to zero volts on the I capacity test, batteries shall be force discharged at 2.0 amperes for a minimum of 5 minutes.

Overload: Batteries shall be discharged at 4.0 amperes.

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Abuse test, predischarge: Designated samples shall be discharged at 2.0 amperes for 1.6 hours.

Abuse test, pulse discharge: Batteries shall be discharged at 2.5 amperes for 1 minute followed by 1.8 amperes for 4 minutes, cycled continuously to cut-off voltage.

Closed-circuit voltage test (battery and cell string): Load resistance of 3 ohms shall be used. Voltage shall be above 6.00 volts within 10 seconds.

Cell closed circuit voltage: Load resistance of 1 ohm shall be used. Voltage shall be above 2.0 volts within 5 seconds.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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